2024 3rd International Conference on Embedded Systems and Artificial Intelligence (ESAI) | 979-8-3315-2205-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/ESAI62891.2024.10913549

Strengthening Security in the Internet of Things (IoT): Integrated Approach of Intrusion Detection Systems (IDS) and Edge Computing

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***Abstract*— The rapid growth of Internet of Things (IoT) applications has led to an increase in sophisticated and malicious attacks, threatening network security and privacy in sectors such as healthcare, industry, and home automation. To address these challenges, The implementation of Intrusion Detection Systems (IDS) has become crucial for detecting and mitigating these threats. At the same time, the emergence of Edge Computing technology offers significant advantages in terms of privacy and performance for IoT infrastructures. However, deploying IDS on resource-limited Edge devices requires innovative solutions to ensure effective protection against cyber threats.**

By integrating IDS on IoT devices and strengthening security at Edge points, it becomes possible to effectively detect and mitigate malicious attacks while optimizing resource usage and ensuring the protection of sensitive data. By placing security at the core of IoT and Edge network design and deployment, it is possible to guarantee a robust and resilient infrastructure capable of addressing both current and future security challenges.

This paper focuses on the study and implementation of machine learning algorithms applied to Intrusion Detection Systems (IDS) within IoT networks. Specifically, we explore the use of real-world data IOT-23 dataset to improve the detection capabilities of IDS. The results show that boosting models, such as XGBoost, CatBoost, and Gradient Boosting, as well as Random Forests, deliver the highest performance with an F1-score of 0.84. These models demonstrate strong capabilities in detecting intrusions while maintaining an ideal balance between precision and recall. These results underscore the effectiveness of these algorithms in improving the security of IoT and Edge networks, paving the way for innovative strategies that allow for optimal resource utilization while ensuring robust protection of sensitive data.

Keywords*— *Internet of Things, Intrusion Detection System, Edge computing, Machine Learning, IoT Security.

I. INTRODUCTION

In the current era, the world is undergoing an unprecedented technological transformation, where the Internet now connects every individual and stands as the central pillar of our society. This evolution is driven by the swift expansion of Internet of Things (IoT) devices, which play a crucial role in facilitating our daily lives across various sectors. In 2024, the global IoT market was valued at approximately 947 $US billion, with forecasts indicating that the number of connected devices will triple by 2030 [1].

According to Statista [2], by 2033, the number of IoT devices worldwide is expected to nearly double, from 18 billion in 2024 to over 32.1 billion in 2030, with a maximum estimate of 35 billion by 2033. China is expected to lead with around 8 billion consumer devices. Major sectors such as electricity, water, retail, and government are rapidly adopting these technologies, with projections of over 8 billion IoT devices across all sectors by 2033.

However, with this increasing connectivity comes a major challenge: security. Data traverses through various layers to reach its destination, exposing devices to risks such as hacking and attacks. Protecting against these threats requires high levels of security, privacy, authentication, and incident resilience, ensuring the sustainability and security of an increasingly interconnected world.

Edge Computing refers to a hybrid approach that combines distributed computing with centralized servers and a significant evolution in modern IT architecture, shifting data processing and storage of IT resources to the network's periphery. Security considerations must include several sub-themes, such as securing information content from being seen or changed, safeguarding functional capabilities against unwanted access, and ensuring regular operations despite maliciously overburdened requests [3].

Securing the edge poses a complex challenge because it requires protecting not only individual devices but also the access points and connections facilitating interaction between these devices and the cloud. Rather than solely securing each device in isolation, it is crucial to implement robust strategies that protect the entire edge network, anticipating various potential entry points for cyberattacks.

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On the other hand, in terms of performance and privacy, Edge Computing technology offers a number of benefits to IoT applications and communication network architecture. However, there are numerous challenges associated with resource limitations when deploying IDS systems on Edge devices. Therefore, IDS solutions based on machine learning have surfaced to tackle these issues [4].

Researchers have studied, analyzed, and summarized the feasibility of securing IoT by focusing on four major areas: blockchain, machine learning, fog computing, and Edge Computing. After evaluating these methods and their resistance to attacks, machine learning emerges as a promising solution when integrated into an end-to-end security strategy, although the addition of firewalls and antivirus remains essential for protecting IT systems [5].

In terms of cybersecurity, detecting intrusion constitutes the crucial first step in effectively protecting our system, and our article proposes a protection solution to ensure security: an Intrusion Detection System (IDS) based on Amazon Echo, Philips Hue devices, and Somfy door locks that uses the IoT23 dataset to identify cyberattacks.

This article's remaining content is separated into four parts. The first addresses related work and foundational information, while the second section proposes the methodology, The section three introduces an implementation step to integrate IDS and develop an intrusion detection model. Lastly, the final section concludes the paper.

II. RELATED WORK

Since the advent of the Internet and connected devices, the latter have become major targets for attacks. The reasons why attackers act and their targets have evolved over time. Protecting IoT devices from botnet attacks and distributed denial of service (DDoS) attack has garnered increasing attention in academic and industrial literature.

Previous research has examined various aspects of detecting and preventing these new threats. Al-Awami and colleagues performed a thorough analysis of IoT architectures, highlighting the crucial importance of security among other key characteristics such as simplicity, cost, and flexibility [6]. Their study underscores current security challenges and explores future trends in the field, particularly the growing use of machine learning for IoT data analysis.

Concurrently, Hussain et al.'s research focuses on advanced techniques for identifying botnet and DDoS attacks in IoT networks. Their two-pronged approach utilizes deep learning models like ResNet to early detect malicious scanning activities and potential DDoS attacks [7]. This approach stands out for its ability to preempt attacks before compromising IoT devices, offering proactive defense against persistent threats.

Furthermore, earlier studies have examined different detection methodologies, such as signature-based techniques and behavioral analysis methods, emphasizing the importance of adaptability and accuracy in detecting evolving botnet attacks [8].

The Internet of Things (IoT) exhibits significant vulnerabilities at each level of its architecture, posing a continuous challenge since its inception [9]. Attacks against IoT devices target multiple layers of their architecture. Recent research has examined various layered architectures proposed for IoT, detailed in Figure 1 by various authors [10]. This figure illustrates how IoT architectures can be structured into three, four, or five distinct layers, each layer employing various technologies that introduce specific security challenges.

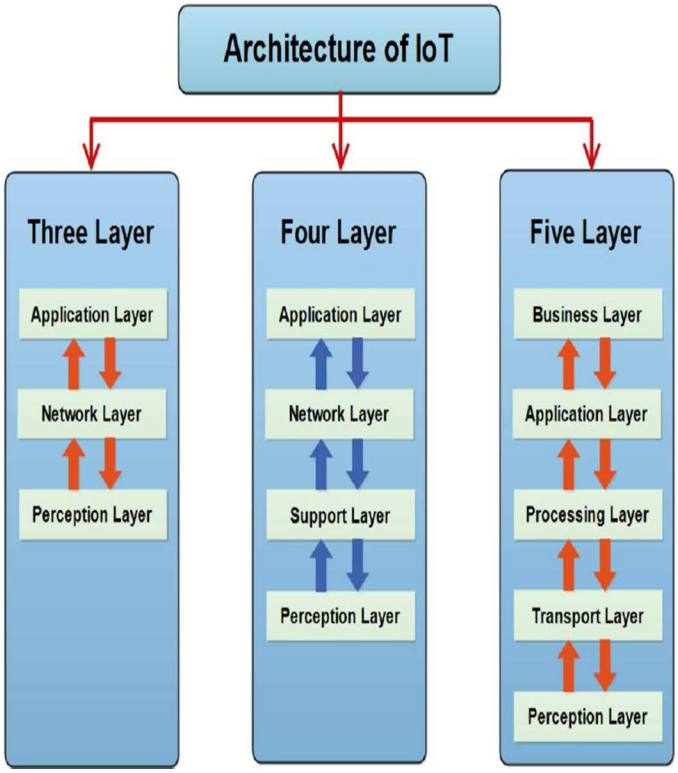


Fig. 1. Architecture of IoT [10]

An extensive study of attack types aimed at the various layers that make up the Internet of Things (IoT) architecture was carried out by Vikas Hassija et al. [11]. Their study categorized and summarized observed attack strategies across the five primary layers of IoT: physical, link, network, transport, and application. They provided detailed insights into each attack type, including potential intrusion vectors, execution methods, and the potential impact on IoT system security (Figure [2]).

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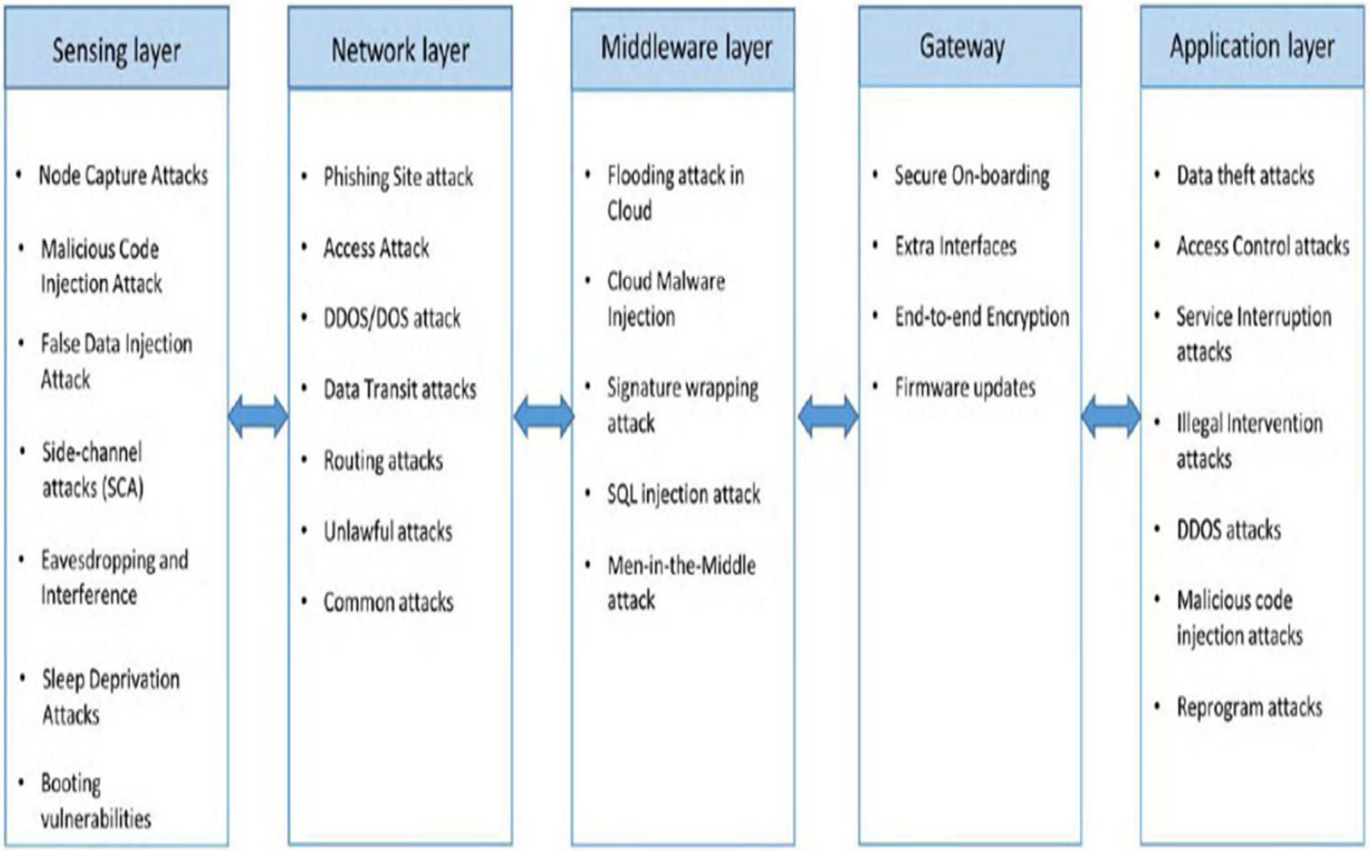


Fig. 2. Types of Attacks on IoT.[11]

Victims of DDoS attacks, according to K. Angrishi (2017), included websites, cloud service providers, individuals, universities, telecommunications companies, and DNS providers like Dyn, who provided services to major platforms such as Spotify, Reddit, Airbnb, Amazon, and others [12].

Das and colleagues [13], as well as Tushir et al. (2021) [14], observed that IoT devices infected by the Mirai botnet are primarily used to launch DDoS attacks against specific targets. Tushir and colleagues studied the impact of these attacks on IoT devices, highlighting an increase of approximately 40% in energy consumption and a doubling of storage usage.

In 2019, Ali Hameed and Alauddin Alomary examined various aspects of security in the Internet of Things (IoT) domain, including advanced authentication, encryption-based attack prevention, and the use of secure hardware techniques such as FPGA-based systems on chips and trusted hardware modules. These studies laid the groundwork for future research aimed at enhancing IoT network resilience against emerging threats [15].

Regarding intrusion detection systems (IDS) specifically in the context of IoT device security, these tools monitor and analyze computer networks and systems to detect and report malicious activities exploiting security vulnerabilities. IDS monitor host or network activities, gather data on these activities, and analyze the data to identify any suspicious or potentially dangerous activity [16]. Network-based intrusion detection systems (NIDS) and host-based intrusion detection systems (HIDS) are the two primary categories into which intrusion detection systems (IDS) are usually divided according to the data source [17]. Anomaly-based and Signature-based detection methods are two commonly used approaches by IDS [18]. In their article, Rafath Samrin and D Vasumathi examine the historical evolution of anomaly-based intrusion detection systems. They trace the origins of this technology since the 1980s, highlighting its post-Internet development to monitor irregular behaviors and abuses on networks. The article emphasizes the introduction of intrusion detection systems (IDS) and host-based intrusion detection systems (HIDS), as well as the impact of projects such as Haystack, Discovery, and Network Audit Director in intrusion identification. These pioneering works laid essential theoretical and technological foundations for contemporary advancements in computer security [19].

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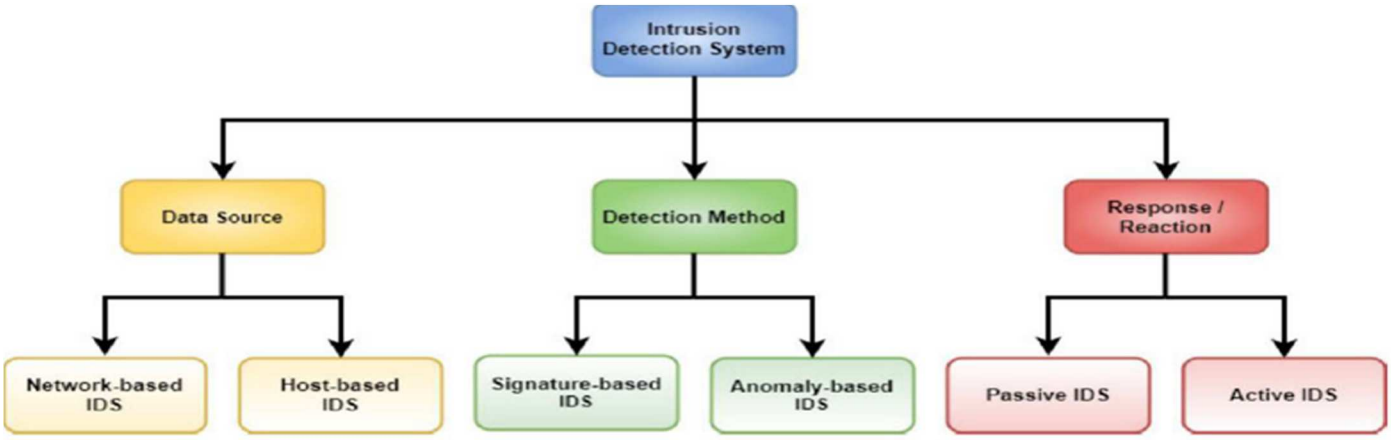


Fig. 3. Intrusion Detection Architecture in IoT [19].

Our article utilized the IoT-23 dataset for investigating machine learning methods capable of automatically detecting and classifying IoT cyberattacks. The principal contribution of this study lies in leveraging the comprehensive IoT-23 dataset to develop an IDS and achieve anomaly detection results.

Similar efforts have been undertaken to develop robust IDS resistant to attacks, utilizing various datasets. Researchers have focused on developing tailored security mechanisms to protect these devices and mitigate threats from both cloud service providers (CSPs) and edge environments [20]. Machine learning approaches have been explored to detect and classify diverse cyberattacks on peripheral IoT networks, evaluating multiple models for their effectiveness [21].

Vishwakarma and Kesswani [22] developed an innovative two-stage intrusion detection system, combining Naïve Bayes machine learning algorithm for data classification and elliptical envelope approach for anomaly detection.

Notable progress has also been made in intrusion detection through the use of various machine learning and deep learning models. Pang et al. investigated the effectiveness of various models, including Generative Adversarial Networks (GANs), in detecting anomalies [23]. Resende et al. explored the use of Random Forest models with diverse datasets for intrusion detection systems [24]. The study by Li and colleagues proposed the use of Convolutional Neural Networks (CNNs) to detect and classify botnet attacks. achieving accuracies up to 86.95% with separate CNN models [25]. Latif and his team developed a Random Deep Neural Network (DRaNN) that achieved an impressive accuracy of 99.54% on the UNSW-NB15 dataset [26]. Xu et al. introduced an LSTM-based autoencoder (AE) model for intrusion detection, surpassing traditional methods with F1 scores up to 99.6% [27]. Finally, The work by Shahriar and colleagues introduced the Generic Intrusion Detection System (G-IDS) framework integrating machine learning-based IDS, demonstrating effectiveness in multiclass classifications within complex environments [28].

III. METHODOLOGY

A. *Data Acquisition and Preparation*

The IoT-23 is a recent database focused on network traffic from IoT (Internet of Things) devices. First released in January 2020, this dataset includes 20 captures of malware executed on IoT devices and 3 captures of network traffic from benign IoT devices. The data was collected between 2018 and 2019 by the Stratosphere Laboratory at CTU University in Prague, Czech Republic, under the AIC (Complex Information Analysis) group of the Faculty of Electrical Engineering (FEL).

The primary goal of IoT-23 is to provide a diverse and annotated dataset on IoT network traffic, including real malware infection cases and examples of benign IoT traffic. The devices used for these captures include Somfy door locks, Philips Hue devices, and Amazon Echo smart assistants, all captured using Raspberry Pi [29].



Fig. 4. Components of the IoT-23 Dataset: Amazon Echo, Philips Hue, and Somfy Door Lock [29].

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In the IoT-23 dataset, after the generation of .pcap files, they were processed using the Zeek network analyzer to create log files. Zeek has the capability to generate connection logs that capture the details of interactions between two entities [30]. The .pcap files were carefully reviewed to extract features associated with specific labels. Following this, a Python script was applied to assign labels to the log files based on the identified patterns. The malicious files varied in size, ranging from a few kilobytes to about 10 gigabytes, with the analysis focusing on individual flows [31].

This dataset includes labels describing the nature of flows associated with potentially harmful activities. These labels were developed at the Stratosphere Laboratory after thorough analysis of malware captures.

Here is an overview of the labels used to identify malicious flows, derived from manual network analysis (Table 1).

TABLE I. OVERVIEW OF MALICIOUS ACTIVITY LABELS AND ASSOCIATED DATA FLOWS IN THE IOT-23 DATASET

Label	Description	Flows	Impact/Explanation
PartOfAHorizontalPortScan	Indicates an activity that appears to be part of a horizontal port scan, a commonly used method to search for vulnerable or open systems on a network.	3,389,036	This activity indicates an attempt to identify vulnerable devices on the network, typically as a precursor to an attack.
Okiru	Refers to a specific type of malware or malicious program, used to infect IoT devices.	1,313,012	This malware specifically targets IoT devices, taking control of them and potentially using them as part of a botnet.
Benign	Indicates a non-malicious or harmless activity or event.	688,812	This type of traffic is harmless and can serve as a reference for identifying network anomalies.
DDoS	Refers to a Distributed Denial of Service attack, where a large number of systems are used to overwhelm the resources of a server or network, making it inaccessible to legitimate users.	638,506	Used to overwhelm a network or server, this attack aims to make services inaccessible to legitimate users.
C&C	Abbreviation for "Command and Control", indicating a connection to a remote server used to control infected systems or botnets.	15,286	This term refers to a communication channel between a server and compromised devices, allowing attackers remote control over them.
C&C-HeartBeat	A periodic connection established between an infected system and its Command and Control server to report its availability or receive instructions.	1,332	This type of connection maintains the link between an infected device and its C&C server, allowing the attacker to check device availability.
Attack	Indicates malicious activity or an attempt to intrude.	538	This label indicates hostile activity aiming to access systems or data without authorization.
C&C-FileDownload	A connection used to download malicious files or payloads from a Command and Control server.	46	This activity downloads malicious files from an attacker-controlled server, propagating malware.
C&C-Torii	Refers to a specific threat or type of malware named "Torii", targeting IoT devices.	30	Torii is malware targeting IoT devices, designed for persistent infection and more challenging detection.
FileDownload	Indicates a file download activity. In this context, it could be used to download malicious files or payloads.	13	This download activity can pose an infection risk if files are unchecked, as they may contain malicious code.
C&C-HeartBeat-FileDownload	A combination of heartbeating connection and file download to a Command and Control server	8	This type of connection combines device status updates with the download of new malicious payloads, reinforcing the infection.
Okiru-Attack	A specific attack involving the use of the Okiru malware.	3	This attack deploys Okiru malware, often in large-scale attacks using IoT botnets.
C&C-Mirai	Refers to a connection or activity associated with the Mirai malware, often used to create IoT botnets.	1	Mirai is known for infecting IoT devices and incorporating them into botnets, commonly used in large-scale DDoS attacks.

B. Data Preprocessing

Before applying modeling techniques, the data underwent several preprocessing steps to ensure their quality and consistency. Here is a table detailing the data columns of our IoT-23 dataset (Table 2).

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TABLE II. DESCRIPTION OF FEATURES

Feature	Description
Uid	Unique identifier for each event or connection.
id.orig_h	Source IP address of the connection.
Id.orig_p	Source port number used.
id.resp_h	Destination IP address of the connection.
id.resp_p	Destination port number used.
Proto	Protocol type used (TCP, UDP, ICMP).
Service	Service or application associated with the connection (DNS, IRC, HTTP, DHCP, SSL, SSH).
Duration	Total duration of the connection.
orig_bytes	Total number of bytes transmitted from the source.
resp_byte	Total number of bytes transmitted from the destination.
conn_state	Current state of the connection (e.g., established, closed).
local_orig	Indicates if the connection was initiated locally (yes/no).
local_resp	Indicates if the response was received locally (yes/no).
missed_bytes	Number of missing or lost bytes during transmission.
History	History of connection state changes.
orig_pkts	Total number of packets transmitted from the source.
orig_ip_bytes	Total number of IP bytes transmitted from the source.
resp_pkts	Total number of packets transmitted from the destination
resp_ip_bytes	Total number of IP bytes transmitted from the destination.

Data cleaning operations included managing missing values by replacing them with the mean of existing values, as well as removing unnecessary variables such as 'local_orig' and 'local_resp', along with other irrelevant metadata for analysis. Features like 'history' were initially excluded due to their sequential nature being irrelevant for classification. We also analyzed the correlation between remaining features such as 'orig_pkts' and 'orig_ip_bytes', as well as 'resp_pkts' and 'resp_ip_bytes', and eliminated highly correlated features to avoid redundancy and reduce the risk of overfitting the models.

C. Encoding and Normalization of Data

Categorical variables like 'proto', 'service', and 'conn_state' were encoded using appropriate methods such as one-hot encoding to preserve contextual information while enabling effective handling by machine learning algorithms. The data was normalized using min-max scaling to ensure all features contribute equally to the modeling process without being biased by their respective scales.

D. Model Selection and Training

For intrusion detection, we have chosen boosting algorithms such as XGBoost, LightGBM, CatBoost, and Gradient Boosting. These algorithms share a fundamental approach of iteratively transforming weak models into strong models. However, they differ in their specific optimization methods and built-in features. For instance, XGBoost stands out for its ability to handle overfitting through advanced regularization techniques, while LightGBM excels in leaf-wise gradient-based tree partitioning efficiency. CatBoost focuses on optimized handling of categorical variables and missing values for enhanced robustness.

On the other hand, Random Forest, although different in approach, distinguishes itself by managing decision tree ensembles in parallel and effectively handling non-linear features and complex interactions. Each algorithm offers unique advantages in terms of performance, scalability, and ease of use, often requiring a choice based on specific data characteristics and modeling goals.

E. Evaluation and Analysis of Results

Once trained, the models were evaluated using standard measures such as the confusion matrix, recall, precision, and F1 score. These measures assessed the capability to accurately differentiate between benign and malicious activities in IoT traffic.

To assess the models' performance, various metrics were used to gauge their accuracy in precise data classification. These metrics included overall accuracy, false positive rate, detection rate, as well as precision (P), recall (Rec), and F1 score. Calculating these metrics involved distinguishing true negatives (TN), true positives (TP), false negatives (FN), and false positives (FP) within the dataset. For example, accuracy is mathematically expressed as follows:

$$Accuracy = (TP+TN)/(TP+FN+TN+FP) \quad (1)$$

Precision represents the proportion of correctly identified attacks out of all packets classified as attacks:

$$Precision = TP/(TP+FP) \quad (2)$$

The F1 score is an overall metric that offers a balanced assessment by taking both recall and precision into account:

$$F1\ Score = 2\times(Precision\times Recall)/(Precision+Recall) \quad (3)$$

Recall (Rec), also referred to as the true positive rate, measures the system's ability to accurately identify attacks during security breaches:

$$Recall = TP/(TP+FN) \quad (4)$$

These measures precisely evaluated how the model distinguished between malicious and benign activities, crucial for securing IoT traffic.

F. Results

In this study, we evaluated seven popular classification algorithms for intrusion detection in IoT traffic: Random Forest, XGBoost, LightGBM, CatBoost, Decision Tree, MLPClassifier, and Gradient Boosting Classifier. The performance of these algorithms was measured on a specific dataset, and the results showed remarkable performance from certain models, particularly the boosting algorithms.

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The Gradient Boosting Classifier outperformed the other methods in terms of overall accuracy, with a weighted average score of 0.86 and a recall of 0.89. It was closely followed by XGBoost and CatBoost, which achieved accuracies of 0.86 and 0.87, respectively. These three models stand out for their robustness in classifying accurately even on complex and diverse datasets. Their main strength lies in their ability to correct prediction errors over iterations, making them particularly effective at detecting subtle and varied malicious behaviors in IoT environments.

In comparison, Random Forest also showed good performance, achieving an accuracy of 0.86 and a recall of 0.89. However, while it is capable of capturing complex relationships between data and handling noisy values, it remains slightly inferior to the boosting models due to the lack of iterative error correction.

Decision Trees and MLPClassifier showed more modest performance. Their inability to effectively handle data complexity and the absence of advanced error correction mechanisms explain their relatively low F1 scores. However, these models are faster and may be considered for applications with lower computational demands, where simplicity and speed are prioritized.

To provide a clearer comparison of The effectiveness of these algorithms across the different metrics (Precision, Recall, and F1 Score), a radar chart has been included. This chart visually represents the strengths and weaknesses of each algorithm.

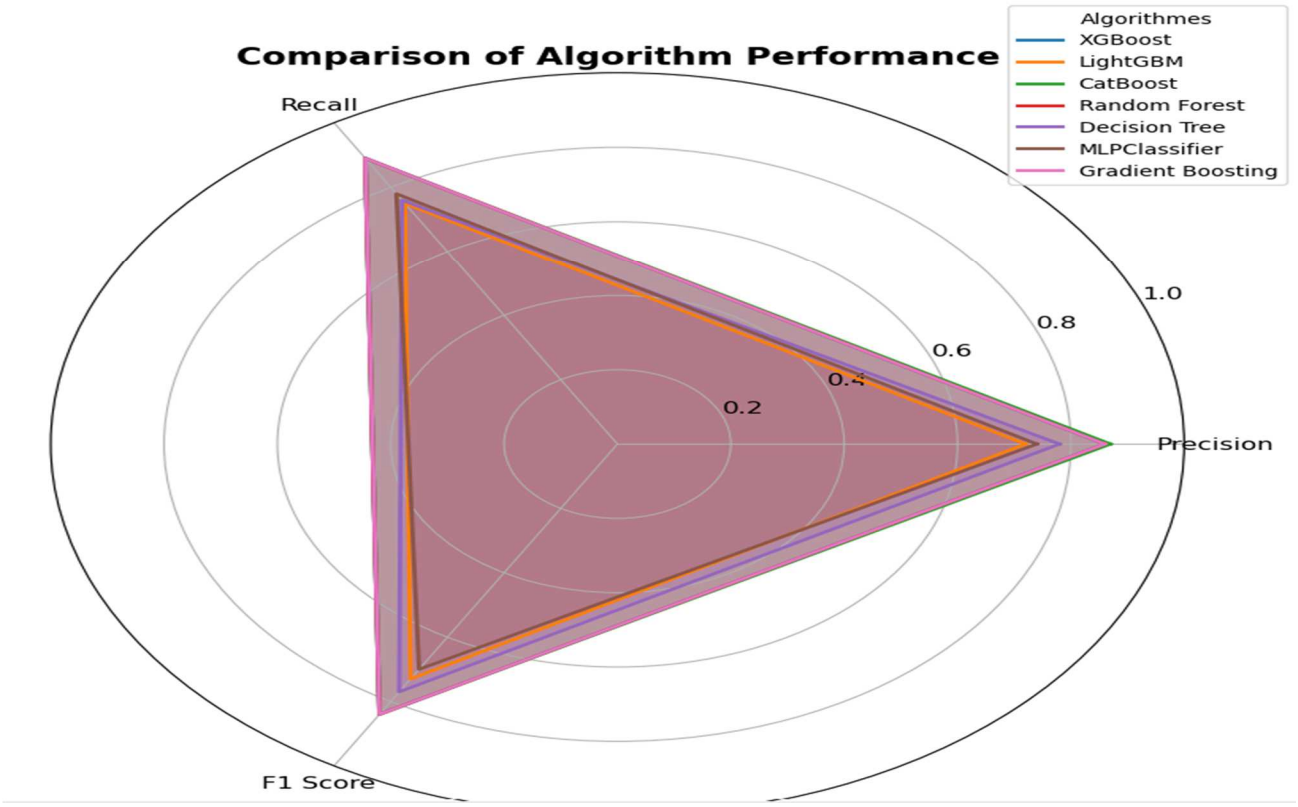


Fig. 5. Radar Chart Comparing Algorithm Performance

As shown in the radar chart, the boosting models (Gradient Boosting, XGBoost, and CatBoost) exhibit superior performance in all three metrics, with Gradient Boosting leading in recall and overall consistency. On the other hand, MLPClassifier and Decision Tree show lower scores, particularly in the F1 score, which reflects their weaker ability to balance precision and recall in this specific IoT traffic dataset.

Additionally, the numerical comparison of the algorithms is presented in Table III, which summarizes the Precision, Recall, and F1 Score for each model.

TABLE III. COMPARISON OF CLASSIFICATION ALGORITHMS FOR IOT SECURITY

Algorithm	Precision	Recall	F1 Score
XGBoost	0.86	0.89	0.84
LightGBM	0.72	0.75	0.73
CatBoost	0.87	0.89	0.84
Random Forest	0.86	0.89	0.84
Decision Tree	0.78	0.76	0.77
MLPClassifier	0.74	0.78	0.70
Gradient Boosting	0.86	0.89	0.84

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IV. Conclusion and Perspectives

In this study, we evaluated several popular classification algorithms for intrusion detection in IoT traffic: Random Forest, XGBoost, LightGBM, CatBoost, Decision Tree, MLPClassifier, and Gradient Boosting Classifier. The results showed that these algorithms are quite effective at distinguishing malicious activities from normal ones. Each algorithm has its strengths: for instance, XGBoost is great at preventing overfitting, CatBoost excels with categorical data, and Random Forest performs well when dealing with non-linear relationships in the data.

However, while these algorithms are powerful for detecting intrusions, It is essential to remember that an Intrusion Detection System (IDS) is just one part of a larger cybersecurity approach. In IoT settings, where devices and networks are becoming increasingly interconnected, IDS needs to be supplemented with other security measures. Technologies like firewalls, antivirus software, and secure communication protocols (e.g., TLS/SSL) are essential in avoiding, detecting, and responding to attacks at different stages. Firewalls can block unauthorized access, antivirus solutions can neutralize malware before it spreads, and secure communication protocols ensure the confidentiality and integrity of data transferred between devices.

To further enhance IDS performance, it's clear that a multi-layered security approach is necessary. Integrating IDS with these complementary technologies would provide a stronger defense against the expanding spectrum of cyber threats. Additionally, while hyperparameter tuning and feature enrichment remain key to improving detection accuracy, integrating advanced data extraction techniques and ensemble learning methods could leverage the combined strengths of multiple models, boosting the overall effectiveness of intrusion detection systems in various IoT environments.

Beyond optimizing IDS themselves, future research could explore how these systems could work in synergy with emerging technologies like blockchain and advanced encryption methods. Blockchain, for example, could offer an immutable record of data exchanges, ensuring data integrity and transparency in IoT networks, while advanced encryption methods could protect sensitive data from manipulation or unwanted access. Studying the performance of IDS in combination with these technologies could open new possibilities for securing IoT systems.

Moreover, applying these models to new datasets and considering real-time learning mechanisms could enhance their adaptability and generalization across a broader range of IoT environments. This would not only make intrusion detection systems more robust, but also more applicable to real-world, dynamic IoT use cases.

In conclusion, while this study has highlighted promising algorithms for intrusion detection, it also emphasizes the necessity of a comprehensive strategy for securing IoT systems. By combining IDS with other layers of security and exploring emerging technologies, we can better address the growing challenges posed by increasingly complex and evolving threats in the IoT landscape.

Acknowledgement

The authors express their gratitude for the support received from the Department of Engineering, Systems, and Applications (ENSA Fès, Morocco). Additionally, the authors would like to acknowledge the support of CNRST under the "PhD-Associate Scholarship - PASS" program.

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